

Project Requirements & Scope Specification

Smart Waste Intelligence Platform (SWIP) • Hackathon Software Demo

1. Executive Summary

Smart Waste Intelligence Platform is an end-to-end municipal and commercial waste management system that combines IoT virtual smart bins, predictive AI dispatch, citizen ticketing, and worker verification.

2. Core Functional Requirements

- **Citizen Portal:** Issue categorization (overflow, missed collection, dumping), photo upload, geolocation, and live ticket tracking.
- **Virtual Smart Bins:** Digital twin simulating fill level %, weight, internal temperature, and battery life with dynamic alert thresholding.
- **AI Priority Engine:** Transparent weighted scoring model evaluating fill urgency, overflow risk, citizen reports, and time since collection.
- **Worker Operations:** Dispatch queue, route assignment, status transitions (Accept -> In Route -> Arrived -> Complete), and verification proof.
- **Admin Command Center:** Unified situational dashboard with real-time KPIs, hotspot visualization, and operational response metrics.

3. Positioning & Transparency

The hardware layer is honestly positioned as a simulated IoT / digital-twin prototype, validating full closed-loop operations without physical hardware constraints.